

# SAHIL PATIL

Mumbai, India

☎ +91 93200 25319 ✉ [sahil90patil@gmail.com](mailto:sahil90patil@gmail.com) 🔗 [linkedin.com/in/byterotom](https://linkedin.com/in/byterotom) 🐙 [github.com/byterotom](https://github.com/byterotom)

## Education

**Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology**

**2022 – Present**

*Bachelor of Technology in Computer Science and Engineering*

*CGPA: 8.51*

**Coursework:** Data Structures & Algorithms, Operating Systems, Distributed Systems, Database Management Systems, Blockchain Technology, Computer Networks

## Coding Achievements

- **Knight on Leetcode with a rating of 1850(top 5%).**
- **Ranked 1048th globally in Leetcode Biweekly Contest 151.**
- **Ranked 1404th globally in Leetcode Biweekly Contest 149.**

## Projects

**Infinity Play** | *Go, PostgreSQL, Cloudflare R2*

**June 2025**

Source Code: [github.com/byterotom/infinity-play](https://github.com/byterotom/infinity-play)

- Developed a full-stack platform in **Go** to play classic HTML5 & Shockwave Flash games on modern browsers.
- Implemented cookie based **JWT** authentication and role-based access for game and admin management.
- Built a server-rendered frontend with **templ** and **HTMX**, using reusable components and **Ruffle** for Flash emulation.
- Used **PostgreSQL** with **sqlc** for type-safe query generation and integrated **Cloudflare R2** with proxy-based access control for securely storing and serving hashed game files.

**Bitwarp** | *Go, Redis, gRPC, HAProxy, RabbitMQ, Docker*

**April 2025**

Source Code: [github.com/byterotom/bitwarp](https://github.com/byterotom/bitwarp)

- Built a distributed **P2P** file-sharing system in **Go** with **gRPC**, enabling **concurrent** chunked uploads and downloads.
- Boosted download speed using multi-source chunking, context cancellation, and **connection pooling**.
- Designed a scalable, fault-tolerant tracker with **HAProxy** load balancing and RabbitMQ-based tracker sync.
- Achieved **100% data integrity** via hashing, verification, and secure chunk-based file reconstruction.

**GoGate** | *Go, Redis, Lua, Apache Bench*

**March 2025**

Source Code: [github.com/byterotom/go-gate](https://github.com/byterotom/go-gate)

- Developed a rate limiter written in **Go** with multiple algorithms (Leaky Bucket, Token Bucket, etc.).
- Leveraged **go routines** and **Redis** for concurrent request handling and validated performance using **Apache Bench**.
- Created a modular, pluggable architecture for seamless integration with proxies, **APIs**, and microservices.
- Improved scalability by offloading state management to Redis with **Lua** scripts, minimizing in-memory footprint and race conditions.

**BlockChat** | *Next.js, Solidity, Ganache, IPFS, Web3Auth*

**January 2025**

Source Code: [github.com/byterotom/block-chat](https://github.com/byterotom/block-chat)

- Engineered an **Ethereum**-based decentralized chat system, ensuring transparency and immutability.
- Deployed **smart contracts** using Hardhat Ignition on a local **Ganache blockchain** for rapid testing and iteration.
- Integrated **Web3Auth** for seamless decentralized identity management through private key reconstruction.
- Enabled secure, real-time communication using **Diffie-Hellman** key exchange and WebSocket-based smart contract event listeners for encrypted chat delivery.

## Technical Skills

**Programming Languages:** Java, Go, C, Solidity, SQL

**Databases & Frameworks:** PostgreSQL, MongoDB, Redis, IPFS, NextJS

**Development Tools & Technologies:** Docker, Git, GitHub, Linux, Nginx, gRPC, RabbitMQ, Cloudflare R2